



CS 212: Design and Analysis of Algorithms

Course Code	CS 212
Course Name	Design and Analysis of Algorithms
L-T-P-C	3-0-2-4
Prerequisites	IC253: Programming and Data Structures, or equivalent for UG students
Intended for	BTech (Semester 3 or 4)
Approval	42nd Senate (Jan 2024), in the B.Tech CSE curriculum; 54th BoA (May 2024)

Course Contents

- Review of Data Structures from IC course: Asymptotic Notations and recurrence relations, Insertion Sort, Merge Sort, QuickSort and Randomization (5 H)
- Sorting Lower Bounds and Non-Comparison Sorting algorithms like Counting Sort and Radix Sort (2 H)
- Order Statistics: Minima and Maxima, Quickselect, Median of Medians (3 H)
- Applying sorting techniques to Computational Geometry : Closest pair problem, Convex Hull of points in a plane (3 H)
- Advanced topics in Sorting and Searching:- Binary Search Trees, Red-Black Trees, Augmentation of Binary Search Trees (8 H)
- Graph Algorithms: Basics of Graphs, BFS and DFS, Topological Sorting, MST(using Kruskal and Prim's), Union-Find Data structure, Single-Source Shortest Path Dijkstra's Algorithm and Bellman-Ford /Maximum Flow (Ford Fulkerson) (8 H)
- Dynamic Programming: Longest Common Subsequence, Matrix Multiplication (2 H)
- Amortized analysis: aggregate analysis, accounting, potential method (3 H)
- Advanced Data Structures: B-Trees, Binomial and Fibonacci/Quake Heaps (3 H)
- Computational complexity: Problem classes: P, NP, NP-complete, NP-hard. Reductions and Examples of NP-complete problems. (3 H)
- Dealing with NP-hard problems:- Approx/Randomized Algorithms, Parameterized Complexity (2 H)
- Coding Lab which covers topics discussed in this course (28 H)

Text Books

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, Introduction to Algorithms, MIT Press, 3/e, 2009.
2. J. Kleinberg and E. Tardos, Algorithm Design, Pearson, 2006.

References

1. S. Dasgupta, C. H. Papadimitriou, U. V. Vazirani, Algorithms, McGraw-Hill, 2006.
2. S. S. Skiena, The Algorithm Design Manual, Springer, 2/e, 2008